Product Specification: VahanaIQ Frontend

1. Product Overview

Product Name: VahanaIQ

Tagline: Sri Lanka Vehicle Finance Intelligence & Lease Affordability Discovery Platform

Description:

VahanaIQ is an interactive web platform that automates Sri Lankan vehicle lease calculations, regulatory down-payment compliance, market catalogue discovery, and deposit savings planning. Designed around Central Bank of Sri Lanka (CBSL) Directions No. 01 of 2026 (clause 2.1), it delivers the capabilities of a financial planning and vehicle comparison workstation — calculating reducing-balance and flat-rate lease rentals, visualising Debt-to-Income (DTI) affordability, comparing curated and AI-extracted vehicles side by side, and planning upfront deposit savings with zero server-side tracking by default.

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2. Goals & Objectives

- **Automate** statutory lease down-payment and Loan-to-Value (LTV) calculations under CBSL Directions No. 01 of 2026 (60% minimum deposit for brand-new, reconditioned, and <1-year registered passenger vehicles; 40% minimum deposit for >1-year registered used vehicles and commercial D-class minibuses).
- **Enhance** buyer financial clarity with real-time Debt-to-Income (DTI) telemetry, interactive SVG speedometer gauges, cashflow spectrum bars, and annual amortisation schedules.
- **Empower** prospective buyers to discover, filter, and compare 227 Sri Lankan vehicle models across 38 makes using sourced pricing, official 8:5 photography, AI-assisted classified ad extraction, and 3-vehicle Head-to-Head matrices.
- **Support** end-to-end purchase preparation through an integrated Deposit & Cash Savings Planner, on-device Catalogue Photo Studio, privacy-safe shareable links, A4 bank-style print quotations, and optional Google account synchronisation.

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3. Target Users

- Individual prospective car buyers and lessees in Sri Lanka evaluating monthly affordability
- Vehicle researchers comparing official distributor models against grey-import and used options
- Household financial planners modelling upfront deposit targets and debt-to-income commitments

4. Core Features

4.1 CBSL Regulatory Lease Calculator & Strategy Presets

Calculates monthly rentals, required deposits, total interest, and total outlay across tenures of 6 to 84 months and interest rates of 0% to 40%. Supports reducing-balance and flat-rate quotes, one-tap strategy presets (Lowest Rental, Fast Payoff, Minimum Cash, 100% Cash), and a live CBSL LTV Split Bar.

4.2 Interactive Budget & DTI Telemetry Console

Visualises borrower affordability in real time using an interactive SVG speedometer dial (0% to 60% DTI) and a stacked Net Income Cashflow Spectrum bar against standard 40% and stretch 50% DTI ceilings.

4.3 Multi-Vehicle Comparison Table & Head-to-Head Finalists

Provides a sortable, filterable comparison table with quick Sri Lankan market segment chips (Under LKR 15M, Hybrid & EV, High Clearance >=170mm, 40% Min Down, Official Agent), inline LKR price editing, inline condition cycling, per-vehicle down-payment overrides, per-vehicle lease detail modals (donut chart and amortisation curve), and a sticky 3-vehicle Side-by-Side Head-to-Head comparison modal.

4.4 Curated Sri Lankan Vehicle Catalogue

Offers a dedicated catalogue (catalogue.html and in-app modal) covering 227 models across 38 makes and 504 official 8:5 gallery images. Labels price provenance (Distributor, Dealer, Launch, Estimated, or Price on Request), supports bulk comparison selection (All, Search, Filtered), and includes personal review bookmarking and export.

4.5 AI-Powered Vehicle Search & Ad Extraction

Integrates with Google Gemini via a serverless proxy to look up single vehicles, resolve multi-vehicle comparative queries, extract prices and conditions from pasted classified ads or uploaded Japanese auction sheet screenshots, and validate Sri Lankan market availability.

4.6 Deposit & Cash Purchase Savings Planner

Calculates forward time-to-goal (months required) and reverse target-date pacing (DD-MM-YYYY) for a vehicle's statutory deposit plus insurance and upfront fees (or full cash purchase). Includes an interactive SVG accumulation trajectory curve, upfront cost composition bar, and named saved scenarios.

4.7 On-Device Catalogue Photo Studio

Provides a browser-based canvas studio (studio.html) to upload custom vehicle photos, frame them to standardized 8:5 proportions (Automatic, Vehicle on a white plate, or Fill the frame), adjust zoom and nudging, set cover order, and preview changes immediately via IndexedDB.

4.8 Unified Cross-Device Sync, Sharing & Statutory Print Quotation

Supports guest-first local persistence, privacy-safe shareable URLs (excluding income and debt unless explicitly ticked), optional Google Sign-In with Firebase Firestore synchronisation and conflict resolution, system/light/dark themes, and an A4 statutory print stylesheet.

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5. Architecture Overview

VahanaIQ follows a modular, zero-build static web architecture paired with serverless cloud endpoints, optimized for instant load times, offline resilience, and strict client-side privacy. The system is composed of the following high-level components:

- **Frontend Web App:** Built with semantic HTML5, CSS3 design-system tokens (css/tokens.css), and native ES6 JavaScript modules across index.html, catalogue.html, studio.html, about.html, privacy.html, and terms.html.
- **Actuarial & Regulatory Engine:** Client-side calculation modules (js/calculator.js, js/savings.js, js/money-input.js) enforcing CBSL Direction No. 01 of 2026 LTV floors, flat-to-reducing rate conversion, and two-decimal LKR currency formatting.
- **Curated Data & Asset Layer:** Static JSON and pre-baked JavaScript data (data/catalogue.json, data/identity.json, js/catalogue-data.js) serving 227 vehicle models, 38 brand marks, and 504 standardized 8:5 gallery assets.
- **On-Device Storage & Offline Layer:** Uses localStorage for loan terms, comparisons, and savings scenarios; IndexedDB for Photo Studio imagery; and a Service Worker (sw.js) for network-first offline caching.
- **Serverless AI Proxy Gateway:** A Cloud Run / Cloudflare Worker service (serverless/) that holds the Gemini API key (gemini-3.6-flash), enforces origin allowlists and rate limits, and routes search and ad extraction requests.
- **Authentication and Cloud Datastore:** Optional Google Identity Services OAuth login coupled with Firebase Firestore (js/account.js, firestore.rules) storing per-user documents (users/{uid}) with pre-write conflict detection.
- **Automated Price Refresh Pipeline:** A monthly GitHub Actions workflow and scraper/validator suite (scripts/scan-prices.mjs, scripts/refresh-prices.mjs) that scans permitted Sri Lankan distributor and importer listings.

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6. Requirements

6.1 Functional Requirements

- Users can configure loan parameters (Rate Type, Interest Rate 0-40%, Tenure 6-84m, Down Payment 40-100%) via inputs, steppers, sliders, or strategy presets.
- The system enforces CBSL minimum deposits (60% for New/Reconditioned/Recent; 40% for Used >1yr and commercial D-class minibuses) and clamps out-of-range inputs with warning hints.
- Users can enter Net Monthly Income, Existing Debt, DTI limit (40% or 50%), Insurance, and Fees to update the SVG DTI speedometer, cashflow bar, and row affordability badges.
- Users can search, filter, sort, and bulk-add vehicles from the 227-model catalogue, requiring a valid positive LKR quote for Price on Request models.
- Users can edit prices inline, cycle conditions inline, override down-payment % per vehicle, inspect Lease Details (donut chart & amortisation curve), and remove rows with Undo.
- Users can pin up to 3 vehicles to the Head-to-Head tray and compare them side by side in the Head-to-Head modal.
- Users can add vehicles via AI Search (single query, comparative query, or ad screenshot upload) or via Manual Entry.
- Users can plan deposit or cash purchase goals in the Savings Planner (by income % or fixed LKR), validate DD-MM-YYYY target dates, and save/load/delete scenarios.
- Users can upload, frame at 8:5, reorder, and export custom vehicle photos in the Photo Studio (studio.html).
- Users can switch themes (System/Light/Dark), generate shareable links, print A4 statutory quotations, and sign in with Google to sync or delete cloud data.

6.2 Non-Functional Requirements

- The system should provide real-time, zero-reload feedback for all user-initiated calculations, sliders, filters, and sort actions.
- Every currency value must display with thousands separators and two decimal places (.00), supporting mid-keystroke comma re-grouping and South Asian numbering.
- All modal dialogs must enforce keyboard focus trapping, inert background isolation, Escape dismissal, and WCAG AA contrast across light and dark themes.
- Client-side privacy must be preserved: no browser API keys, local-only IndexedDB for studio photos, and opt-in income sharing.